

USBSID-Player manual

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Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give **no guarantees** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

What!?

Yes you read that right, USBSID-Player is embedded on USBSID-Pico and allows you to play PSID, RSID, PRG or P00 chiptunes without the need of an emulator on your computer!

Firmwares with embedded USBSID-Player

Starting with firmware version v0.6.0-BETA firmwares ending with `sp` have an embedded emulator that can play PSID and RSID tunes and PSID64 PRG/P00 files.

For example `usbsidpico2-v1.0_sp.uf2`, `usbsidpico2-v1.3-rgb_sp.uf2` or `usbsidpico2-v1.5_sp.uf2`

This USBSID-Player is still in active development and some tunes might not work or play at all.

The embedded USBSID-Player requires more than 256KB RAM, which is why it is primarily built and tested for rp2350 Pico2 boards, but `_sp` firmwares are also built for rp2040 Pico1 boards.

NOTE! Due to the hardware limitations of the RP2040, highly demanding, cycle-exact SID tunes (dense per-cycle register writes, e.g. some digi/sample tunes) may not play at full speed on Pico1 or Pico1_w boards. Use a Pico2 or Pico2_w (rp2350) board for these tunes.

WARNING! Whenever the USBSID-Player is used the Pico MCU is overclocked to 250MHz, regardless of Pico type. Passive cooling *might/could* be needed.

Cynthcart runs through the embedded USBSID-Player - see the `USBSID-Pico-Cynthcart-manual` file for more information.

Usage

To play SID tunes and run PRG/P00 PSID64 files *on* USBSID-Pico you will need the `send_sid` tool that is available in the `tools.zip` starting with firmware release v0.6.0-BETA and up.

The help menu is self explanatory.

Multi-subtune PSID/RSID files are supported - you can switch to the next or previous subtune while a tune is playing.

Future plans

When all bugs are squashed and more tunes are supported a website will be made available where you can 'send' files from to the onboard SID player

Development

More information about USBSID-Player is available on the [github repo](#)

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